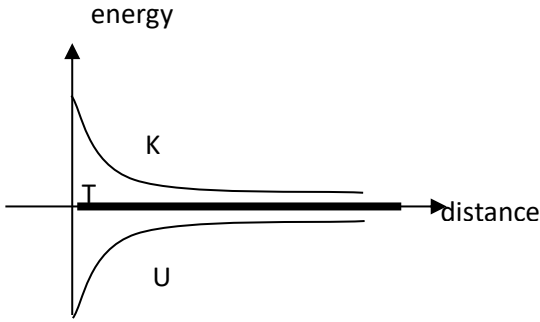


2	(a)	(i)	$KE_i + PE_i = KE_f + PE_f$ $\frac{1}{2}mv^2 + \left(-\frac{GMm}{R_E}\right) = 0 + \left(-\frac{GMm}{2R_E}\right)$ $v = \sqrt{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{6.38 \times 10^6}}$ $= 7.90 \times 10^3 \text{ m s}^{-1}$	M1 A1
		(ii)	1. 0 J	A1
			2. Total energy = 0 J By conservation of energy, all KE is converted to GPE, hence KE graph is a reflection of GPE graph  One mark for shape One mark for positive values	A2
	(b)	(i)	$\frac{GMm}{r^2} = mr\omega^2$ $\frac{GM}{r^3} = \left(\frac{2\pi}{24 \times 3600}\right)^2$ $r = 4.22 \times 10^7 \text{ m}$ Hence, $h = r - 6400000 = 3.58 \times 10^7 \text{ m}$	M1 A1 M1
		(ii)	$v = r(2\pi)/T$ $= (4.22 \times 10^7)(2\pi) / (24 \times 60 \times 60)$ $= 3068.9 \approx 3070 \text{ m s}^{-1}$	M1 A1

		(ii) $E_T = E_k + E_p = \frac{GMm}{2r} + \left(-\frac{GMm}{r} \right) = -\frac{GMm}{2r}$ $= \frac{-6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 1000}{2 \times 4.22 \times 10^7}$ $= -4.74 \times 10^9 \text{ J}$ Alternatively, $U + E_k = \frac{-GMm}{r} + \frac{1}{2}mv^2$ $= \frac{-6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 1000}{4.22 \times 10^7} + \frac{1}{2}(1000)(3068.9)^2$ $= -4.77 \times 10^9 \text{ J}$	M1 A1
		(iv) The satellite will have a smaller gravitational energy and larger kinetic energy.	B1